Working Elderly in Japan

Sagiri Kitao*
Nozomi Takeda†

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Abstract

In this paper, we first document trends in the Japanese labor market over the past few decades, with a particular focus on the elderly. Employment rates among the elderly in Japan are relatively high compared to other OECD countries and have increased sharply over the last two decades. We also examine the characteristics of elderly workers, including their industry distribution, health status, and wealth. We then develop a structural life-cycle model with consumption-saving decisions and endogenous labor supply in both intensive and extensive margins to evaluate the factors influencing employment. We find that social security reform to raise the retirement age by five years and reduce the replacement rate by 20% would increase the participation of men in their 60s from 58% to 69% and 67%, respectively, while also encouraging higher saving for retirement. Additionally, overall labor productivity growth will reduce elderly participation, and productivity growth among the elderly, driven by low skill depreciation, will motivate them to stay in the labor force longer.

Keywords: Retirement, life-cycle model, health, public pension, Japan.
JEL Classification: D15, E21, E24, H55, J14,

*National Graduate Institute for Policy Studies. Email: sagiri.kitao@gmail.com
†University of Tokyo, Email: takedanozomi@g.ecc.u-tokyo.ac.jp

1 Introduction

Ongoing demographic aging in many countries presents a significant economic challenge, primarily due to the shrinking labor force and the rising fiscal pressures from age-related public expenditures. The labor supply, however, is influenced not only by the size of the working-age population but also by the participation rates across individuals of different age groups and their labor productivity.

This paper consists of two parts. In the first part, we examine labor market trend in Japan and present a descriptive empirical analysis, with a focus on the behavior of different age groups, placing particular emphasis on the elderly. Using multiple data sources, including the Basic Survey of Wage Structure (BSWS), Labour Force Survey (LFS), and Japan Household Panel Survey (JHPS/KHPS), we analyze the life-cycle employment and earnings pattern over time since the 1990s and across different cohorts.

Employment rates of elderly men and women across age groups are high relative to other developed countries and have been on the rise since 2000. Men's employment rate in their early 60s rose from 65.1% in 2000 to 84.4% in 2023, and women's from 37.8% to 63.8%. Earnings are hump-shaped over the life-cycle age and the trend differs from that of employment rates. Men's earnings at young and middle ages have declined since 1990s and very recent cohorts of men earn less than those in earlier cohorts. We study personal characteristics of working elderly and find that health, marital status, receipt of public pensions, and the level of household assets appear to matter for employment status of men. The industry distribution of elderly men and women also shifted significantly over the last two decades, since the early 2000s. The share of those working in the agriculture, forestry and fisheries declined sharply from 25% to 12% for men and from 30% to 10$ for women. Instead, the shares for workers in the services, medical and health care services increased to absorb the share.

In the second part, we develop a structural life-cycle model, calibrated to the economic environment and labor market experiences of male individuals born between 1936 and 1940. This cohort was chosen to allow us to track the full career trajectory of the same group of individuals.

We simulate the model under alternative assumptions about the social security scheme and the labor productivity of young and old workers. Our findings indicate that the specifics of social security rules significantly impact individuals' labor supply decision. Raising the retirement age from 60 to 65 and reducing the replacement rate by a 20% would increase the average employment rate of men in their 60s from 58.1% to 74.9% in total. Individuals also accumulate more wealth over the life-cycle, as a result of optimally choosing the combination of additional labor supply and more savings to prepare for retirement.

While higher overall productivity tends to reduce elderly participation due to the

income effect, an increase in the productivity of elderly workers (or a decrease in the rate of skill depreciation) leads to higher participation among them. While economic growth might generally lower participation rates, improving working conditions for the elderly could counteract these negative effects and enhance their employment.

This paper builds on the literature that investigates the life-cycle pattern of labor supply and its interaction with macroeconomic environment, using a quantitative model of heterogeneous individuals. French (2005) is an early paper that estimates a life-cycle model with endogenous saving and labor supply in both intensive and extensive margins and studies the roles of social security scheme on individuals' decisions. French and Jones (2011) build a life-cycle model of older individuals with medical expenditure risks and show that Medicare plays an important role in retirement decisions of the elderly. Fan et al. (2024) construct a model with endogenous consumption, labor supply and human capital and emphasize the roles of human capital accumulation and depreciation in accounting for the life-cycle profiles of labor supply and wages. Borella et al. (2023) examine effects of marriage-related tax provisions and social security benefits and demonstrate distortionary effects of these provisions on saving and labor supply of men and women.

There are papers that focus on the roles of health on life-cycle behavior of households. De Nardi et al. (2024) investigates the lifetime costs of bad health, focusing on how predetermined health types influence economic outcomes over the life-cycle. Capatina (2015) investigates how health risks affect labor supply, asset accumulation, and welfare over the life-cycle through different channels including productivity, medical expenditures, time endowment, and survival probabilities.

Papers listed above parametrize the models to the U.S. economy. There are recent papers that build quantitative life-cycle models focusing on the Japanese economy and study roles of various policies on life-cycle decisions of households.[1] Braun and Joines (2015) analyze the medium to long-run impact of demographic aging and fiscal sustainability in a general equilibrium life-cycle model. Kitao (2015) builds a model with endogenous labor supply in both intensive and extensive margins and studies the roles of demographic aging and the effects of social security reform.

Some papers are focused on the roles of fiscal policy on labor supply decision of women. Yamada (2011) studies the impact of tax reform in teh 1990s on labor supply decisions of married women over the life-cycle. Kitao and Mikoshiba (2024) construct a model of labor supply and human capital accumulation of women with heterogeneity in marital status and family structure. Okada (2023) builds a life-cycle model of the elderly and studies the effects of the social security's earnings test on their labor supply, saving and

[1]For quantitative models of labor supply in other countries, see, for example, Laun and Wallenius (2015), who examine the impact of the Swedish pension reform, focusing on how the reform affects retirement behavior and disability insurance claiming of old workers. Alonso-Ortiz (2014) uses a life-cycle model with incomplete markets to study the roles of social security features to account for the variation in employment rates across OECD countries.

welfare.

Our study is also related to the empirical literature on the labor supply of the elderly in Japan. Oshio et al. (2020) and Oshio et al. (2023) investigate the effects of public pension programs and implicit taxes on work incentives of the elderly. Oshio et al. (2024) analyze old workers' health capacity to work using data on life expectancy and health status of young and old workers, and find that the elderly in Japan have significant health capacity to continue working. Kondo (2016) studies whether a rise in elderly employment crowds out young workers, finding no clear evidence of substitution, though there is some indication of a modest crowding out of middle-age female part-time workers.

2 Data and Policy: Elderly's Work Environment

2.1 Data Source

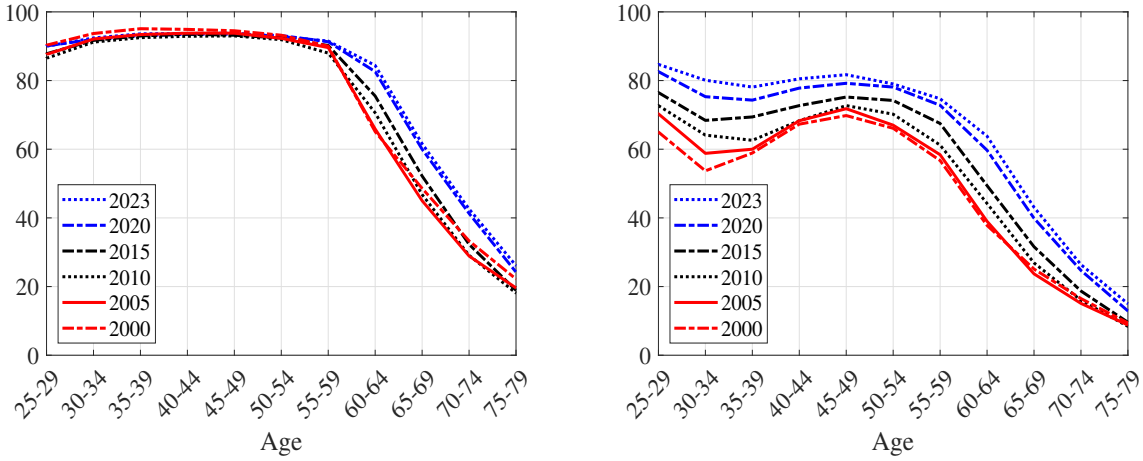
In this section, we study the trend of labor market participation over the life-cycle as well as earnings and work hours over the last few decades. For employment statistics, we use the statistics based on the Labour Force Survey (LFS) conducted by the Statistics Bureau of the Ministry of Internal Affairs and Communications (MIC). The survey has been conducted every month since 1947 and collects comprehensive information on the current state of employment and unemployment of individuals residing in Japan.

For the information about wages and work hours, we use the statistics from the Basic Survey of Wage Structure (BSWS). The BSWS is an establishment survey conducted annually by the Ministry of Health, Labour and Welfare, since 1947. Both the LFS and the BSWS are the Fundamental Statistics under the Statistics Act.

We also use the micro data of the Japan Household Panel Survey (JHPS/KHPS) to investigate various factors related to participation of the elderly. It is conducted by the Panel Data Research Center at Keio University since 2004 and collects comprehensive information on 4,000 households and 7,000 individuals nationwide. It includes questions regarding various aspects of households, including family structure, employment of family members, health status, assets, etc.

2.2 Labor Market Trends

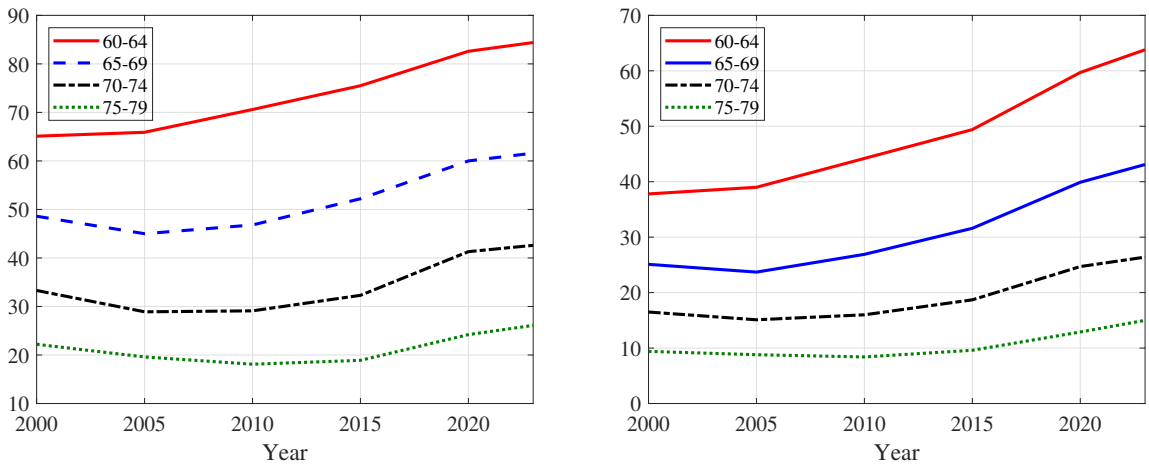
Figure 1 shows the recent trend in the participation rates of men and women, in different years between 2000 and 2020. There is little change in the participation rates of prime-age men between age 20 and 50, more elderly men participate in the labor market in recent years. For women, participation rates increased across all age groups.



(a) Men (b) Women

Figure 1: Employment Rates of Men and Women by Year (Source: Labour Force Survey)

Figure 2 shows participation rates by selected age groups to highlight the trend among the elderly population.

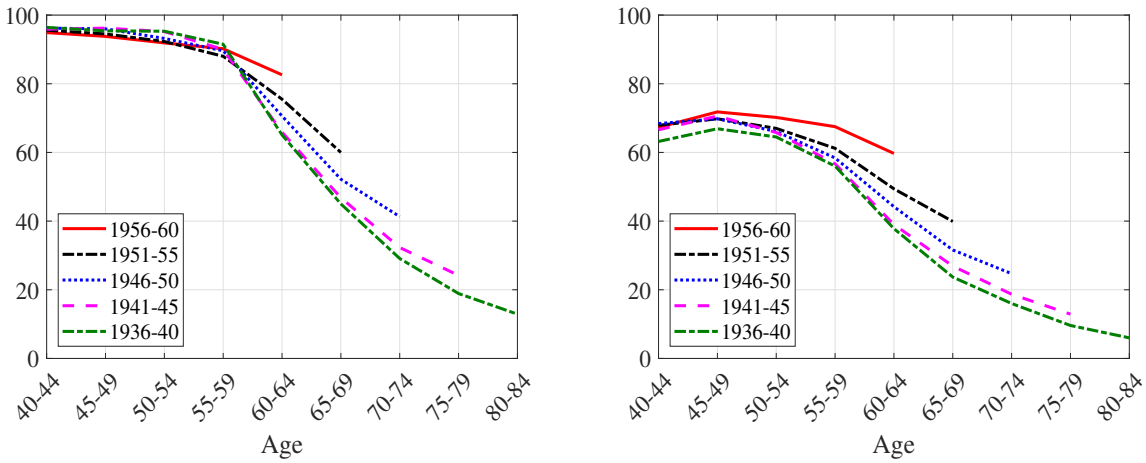


(a) Men (b) Women

Figure 2: Employment Rates of Men and Women by Age Group (Source: Labour Force Survey)

Figure 3 shows the trend of the elderly's participation from a cohort perspective. Using the cross sectional data on labor force participation by age, we constructed the life-cycle profiles of participation for cohorts born between 1936 and 1960. More recent cohort stay in the labor force longer and many of them continue to work until after their 60s. The participation rate of men aged 60-64 is above 70% for all cohorts and it increased from slightly above 70% for the early-1940s cohort to above 85% for the late-1950s cohort. More

than 40% of the late-1940s cohort aged 70-74 now are still working. The participation of elderly women shows a similar trend, though the participation rates are lower than men.



(a) Men (b) Women

Figure 3: Employment Rates of Men and Women by Cohort (Source: Labour Force Survey)

There has been much attention to the participation rates of women, and especially how the M-shaped pattern has diminished during recent decades as more women stay in the labor force in their late 20s and 30s.

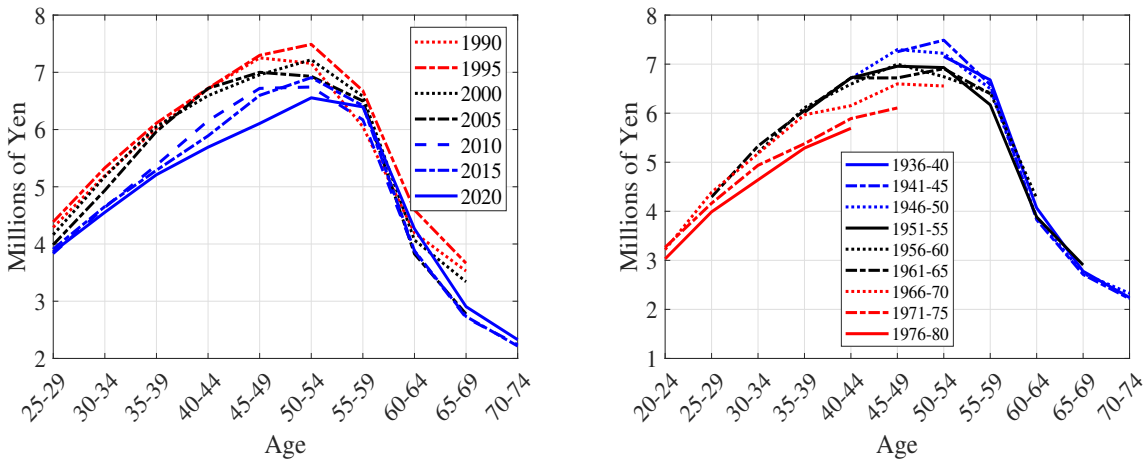
In many quantitative models that focus on women's labor supply, men's participation decisions have received less attention and often times models abstract from them entirely, by assuming that men always work. Many models also assume that individual leave the labor force and collect social security benefits at some exogenous age, typically the eligibility age of 65 for the pension.

The figures above, however, demonstrate many men continue to work well after the normal retirement age of public pension. Moreover, there has been a major increase in the labor supply of elderly men in recent decades. This paper focuses on men's life-cycle work decisions, tries to understand the factors that influence participation decisions of elderly men and what might be behind the rise in participation during the last few decades. We focus on behavior of men belonging to different cohorts.

Figure 4 shows the average annual earnings of male workers by year and by cohort computed from the Basic Survey on Wage Structure (BSWS). Although not entirely monotone across age groups and over years, there is a visible declining trend of earnings over the last few decades among prime-age men, as shown in Figure 4a. For example, average earnings of men aged 40 fell by 14% from 6.7 million yen in 1990 to 5.7 million yen in 2020.

Figure 4b shows the age-profile of earnings by cohort, computed from the cross sec-

tional average, as we did in computing cohort participation rates above. This also show a decline in earnings throughout the life-cycle for more recent cohorts.



(a) By Year

(b) By Cohort

Figure 4: Earnings of Male Workers (Source: Basic Survey on Wage Structure)

Figure 5 shows the average work hours of male workers across age groups in 1990-2015.[2] Work hours declined sharply across age groups in the 1990s. In the early 1990s, Labor Standard Act was revised, lowering weekly work hours from 48 hours to 40 hours in principle, with some exceptions. In each year, average work hours are flat between 25 and 59 years old and they fall sharply thereafter.

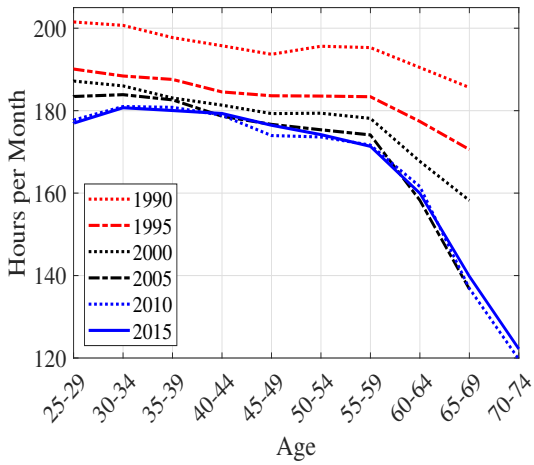
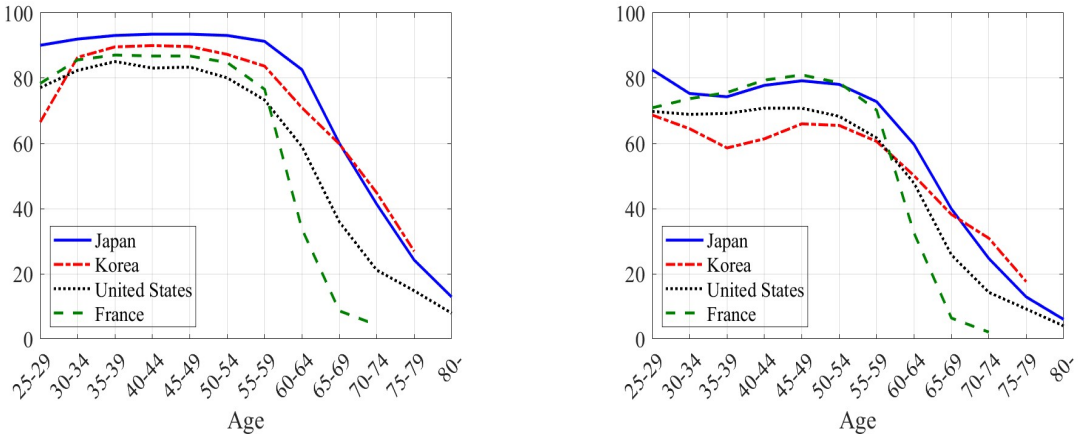


Figure 5: Work Hours of Male Workers (Source: Basic Survey on Wage Structure)

[2]We intentionally focus on this time period, not including data after 2020. There was a change in the workers included in the category of short-time workers in 2020 and we are not able to compute the average work hours under a consistent definition.

2.3 Labor Market Trends in Other Countries

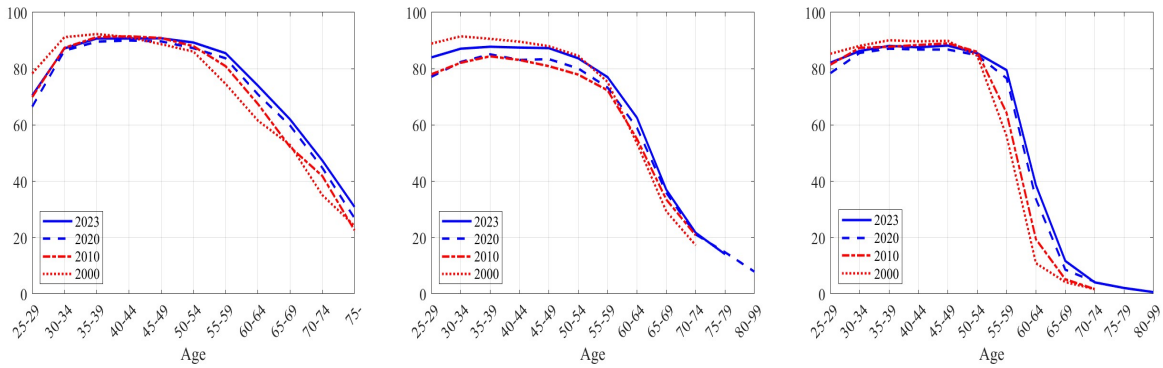
Are the relatively high participation of the elderly and the trend or rising employment rates of the elderly unique in Japan? Figure 6 shows employment rates of women and women in other countries including Korea, U.S. and France.



(a) Men (b) Women

Figure 6: Employment Rates of Other Countries in 2020 (Source: OECD)

Figure 7 shows the trend of male employment rates in Korea, U.S. and France. Although the levels differ across countries, employment rates of the elderly above age 60 are rising in all three countries, following the similar trend to that of Japan. Although the quantitative analysis of this paper focuses on the Japanese economy, the methodology is likely applicable to analysis of the labor market trends in other countries.



(a) Korea (b) United States (c) France

Figure 7: Employment Rates of Other Countries by Year (Source: OECD)

2.4 Factors Related to Elderly Participation in Japan

We now use the micro data from the Japan Household Panel Survey (JHPS/KHPS) to investigate factors related to participation of elderly males. We first study the relationship between labor force participation and personal characteristics by running the following probit regression:

$$P(y_i = 1 \mid j \geq 60) = \Phi(\alpha + \boldsymbol{x}_i\beta + \varepsilon_i), \tag{1}$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution, and the dependent variable y_i takes the value 1 if the individual aged 60 and above is employed, and 0 otherwise. Explanatory variables $\boldsymbol{x}_i$ include age; the square of age; an indicator of fair health status; an indicator of poor health status; an indicator of high education[3]; an indicator of being married; log of asset; an indicator of whether the individual receiving a pension. ε_i is the error term.

Table 1 presents the results of the probit regression, displaying the coefficients in the first column and the corresponding average marginal effects in the second. The survey asks respondents to report their health status by choosing one of five options, listed in order of descending health conditions. We group the first three responses as "good," the fourth response as "fair," and the last one as "poor," comprising the three health statuses. Health status emerges as a crucial determinant of labor force participation among the elderly. The results reveal a clear negative association, with deteriorating health leading to a decreased likelihood of labor force participation. Compared to individuals reporting good health, which is the reference category, those in fair health have a 10.2 percentage point lower probability of being employed, while those in poor health have a substantial 30.6 percentage point lower probability. These results underscore the critical role of health status in determining the ability and willingness of older individuals to remain in the workforce.

[3]High education includes those with a college degree or higher, while those with lower educational levels are classified as low education.

Table 1: Personal Characteristics and Labor Participation of Elderly

Variable	Coefficient	Average Marginal effects
Age	$-.221^{***}$	$-.071^{***}$
	(.042)	(.014)
Age^2	$.001^{***}$	$.0003^{***}$
	(.0003)	(.0001)
Health		
Fair	$-.316^{***}$	$-.102^{***}$
	(.031)	(.010)
Poor	-1.036^{***}	$-.306^{***}$
	(.085)	(.021)
Education	.0126	.004
	(.026)	(.008)
Marital Status	$.217^{***}$	$.069^{***}$
	(.035)	(.011)
Pension	$-.687^{***}$	$-.228^{***}$
	(.038)	(.012)
Log Asset at Age 60	$-.042^{***}$	$-.013^{***}$
	(.004)	(.001)
Constant	11.208^{***}	-
	(1.52)	-

Note: (*), (**), (***) denote significance at the 10%, 5%, and 1% levels, respectively. Sample period: 2004–2022. Number of observations: 13,471. Pseudo R^2: 0.1731. The sample includes individuals aged 60 or older. Values in parentheses represent the standard errors for coefficients and the Delta-method standard errors for average marginal effects. For the log of assets at age 60, if the asset at age 60 is not reported, we use the earliest reported asset value after age 60 as a proxy.

In addition to health, other personal characteristics also play a role. Age exhibits a non-linear relationship, with the probability of working decreasing with age. Education shows a positive but not statistically significant association, suggesting a potential link between higher education and continued working. Marital status and gender are both significant, with married individuals and men being more likely to participate in the labor force. Pension receipt is associated with a lower probability of employment, likely reflecting the financial security provided by pensions that enables some individuals to retire. Finally, the log of assets has negative coefficient, indicating that wealthier individuals may be less inclined to work.

Health: As shown in the regression analysis above, health is an important factor in accounting for the labor force participation over the life-cycle. Using the JHPS/KHPS data, we divide individuals into three health status, good, fair and poor, according to their self-reported condition of health.[4] Figure 8 shows the shares of individuals in fair and poor health status. Note that we use male samples of [YEAR]. The health status deteriorates as individuals age, and the share of those in fair and poor health increases from less than 10% in their 20s to 20% in their 70s.

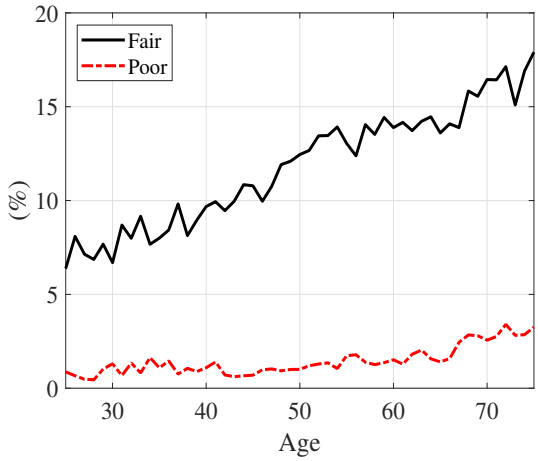
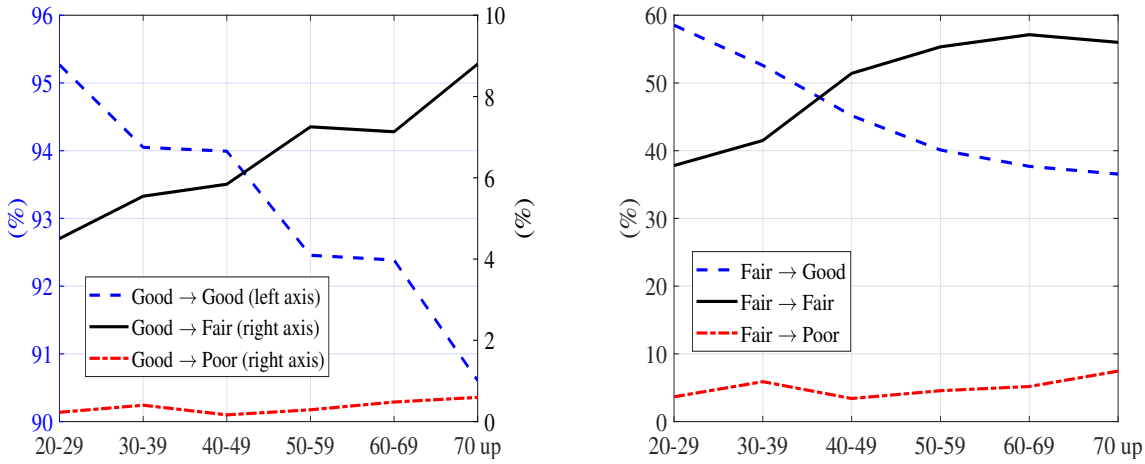


Figure 8: Shares of Individuals in Fair and Poor Health Status (Source: JHPS/KHPS)

Figure 9 report the transition probabilities of health status for individuals in good and fair health status. As shown in Figure 9a, the probability of staying in good health status declines in age and more individuals transit to either fair or poor health status. From fair health status, fewer individuals recover to good health status as they age and more individuals remain in fair health status or experience a further deterioration to poor health, as shown in Figure 9b. These movements contribute to a rise in the share of individuals in fair and poor health status.

[4]The survey asks individuals to report their health status and to choose one of five options. We combine the first three answers as good and the remaining two as fair and poor.



(a) From Good Health

(b) From Fair Health

Figure 9: Transition of Health Status (Source: JHPS/KHPS)

As indicated in the regression analysis, health plays an important role in accounting for the participation. Participation rates of individuals in poor health status are significantly lower than those in good or fair health. Since the share of individuals in fair and poor health status increases in age, deterioration of health contributes to a more important role of health in participation decision of old individuals. We include health as one of state variables and model the transition process of an individual's health status over the life-cycle in the quantitative model presented in Section 3.

2.5 Fiscal Policy, Public Pension, Health Insurance and Long-term Care

In this section we provide a brief review of social security and fiscal systems in Japan surrounding workers and their life-time budget. In Japan, the pension system started as workers' insurance scheme in 1942, and the national scheme started to provide a universal coverage in 1961. The public pension system consists of two parts, the national pension system that provides basic pension benefits, and employees' pension insurance system. The benefits of the second part are determined based on the contributions made at work. The standard eligibility age to receive the basic pension benefits has been 65, and the eligibility age for the employees' pension has been increasing from 60 since 2000, and will reach 65 in 2025.

The public pension, health insurance and long-term insurance, constitute the three main pillars of the social security system in Japan, accounting for 20% of GDP in 2021, according to the Financial Statistics of Social Security in Japan reported by the IPSS. Figure 11 shows the expenditures of the three programs as a share of GDP. Note that the long-term care insurance started in 2000.

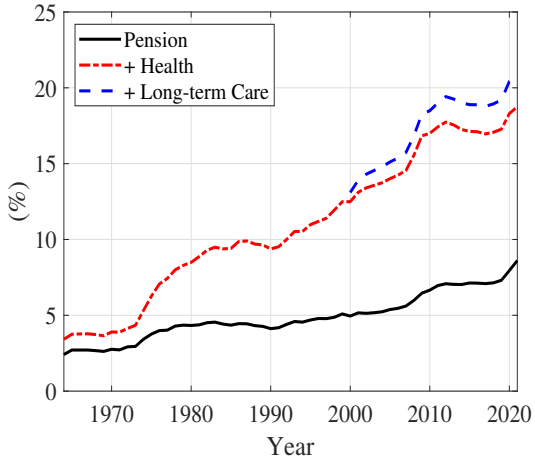
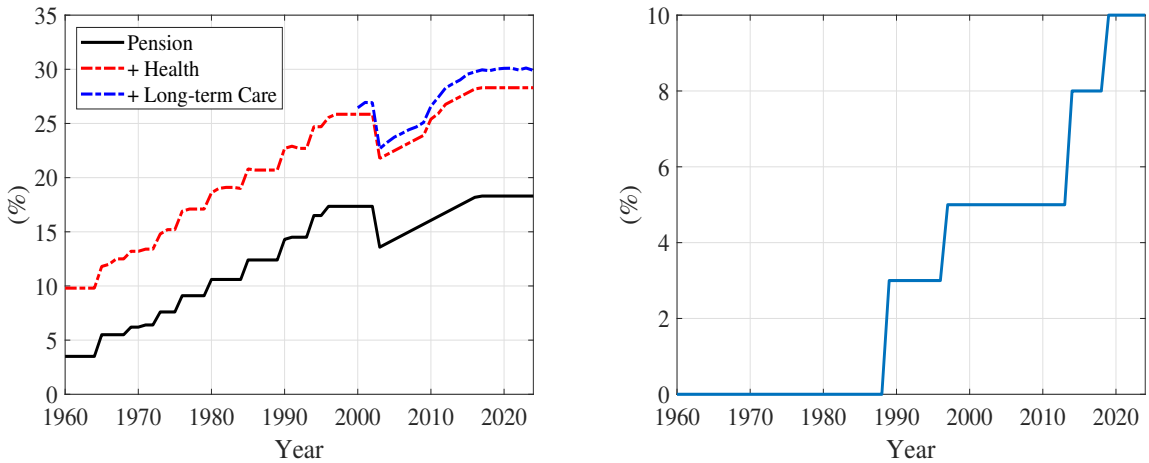


Figure 10: Social Security Expenditures as a Share of GDP (Source: IPSS)

With a steady increase in the expenditures, the tax burden has increased over time. Figure 11 shows the path of social security taxes for each program. The consumption tax was introduced in the late 1990s and rose to 10%.



(a) Social Security Tax

(b) Consumption Tax

Figure 11: Social Security and Consumption Tax

2.6 Employment Across Industries

We now examine the distribution of workers across industries, by age group and over time. We use the LFS data since 2003 and summarize the share of workers in selected industry groups in Table 2.[5] The LFS classifies industries into 20 categories according to the Japan Standard Industrial Classification. We further consolidate them into 7 larger

[5]Industry categories vary over time in the LFS and we chose the starting year of 2003, in which there were major changes in the industry group.

categories.[6]

Table 2: The Share of Each Industry Among the Employed

	Agriculture	Manufacture	Services	Medical	Transport	Wholesale	Others
Male							
≥ 65							
2003	25.3	23.3	25.7	3.7	4.7	15.5	1.4
2013	16.5	23.4	30.6	4.6	7.2	14.4	2.6
2023	12.2	23.4	33.7	6.7	7.5	12.4	3.9
≤ 64							
2003	2.7	36.1	24.7	3.1	11.2	15.3	6.9
2013	2.4	33.2	25.5	5.0	12.1	14.3	7.4
2023	1.9	31.9	25.4	6.1	13.5	13.5	7.5
Female							
≥ 65							
2003	30.0	14.4	28.3	5.0	0.6	20.0	1.1
2013	16.6	13.0	38.1	10.5	1.6	18.2	2.0
2023	10.3	11.6	36.6	18.7	2.1	17.4	3.4
≤ 64							
2003	3.0	18.9	33.6	15.6	4.0	21.8	3.1
2013	2.0	14.1	33.6	21.6	4.4	20.1	4.0
2023	1.3	13.6	33.6	22.8	5.7	18.0	5.0

Source: Labour Force Survey

Note: The units are percentages. All industries are covered, but due to rounding, the sum of each row may not equal 100.

The data shows a noticeable shift in employment trends among males aged 65 and over from 2003 to 2023. The share of those employed in agriculture has significantly decreased from 25.3% in 2003 to 12.2% in 2023. There has been an increase in the share of male

[6]The 7 categories are: Agriculture (including Agriculture and Forestry and Fisheries), Manufacturing (including Mining and Quarrying of Stone and Gravel, Construction, and Manufacturing), Services (including Finance and Insurance, Real Estate and Goods Rental and Leasing, Scientific Research, Professional and Technical Services, Accommodations, Eating and Drinking Services, Living-Related and Personal Services and Amusement Services, Education, Learning Support, Compound Services, and Services, N.E.C.), Medical (including Medical, Health Care and Welfare), Transport (including Transport and Postal Activities, and Information and Communications), Wholesale (including Wholesale and Retail Trade), and Others (including Electricity, Gas, Heat Supply and Water, Government, Except Elsewhere Classified, and Industries Unable to Classify).

workers in the services sector, from 25.7% in 2003 to 33.7% in 2023, and the share of the medical/healthcare sector also increased by 3 percentage points.

By comparing these trend with males aged 64 and under, a similar but less pronounced pattern emerges. The share of young and middle-aged male workers in agriculture also decreased from 2.7% to 1.9%, while the medical/healthcare sector saw an increase, from 3.1% to 6.1%. This suggests that while both older and younger males are moving into services and medical/healthcare sectors, younger males experiencing a more moderate shift.

Female workers aged 65 and over also shows notable changes in the employment patterns across industries. The share of agriculture drastically decreased from 30.0% in 2003 to 10.3% in 2023, a steeper decline than that observed for men. The most substantial increases are in the services and medical/healthcare sectors. The share of elderly women in services increased from 28.3% to 36.6% and that in medical/healthcare sector rose from 5.0% to 18.7%. This observation is consistent with the findings of Kawaguchi et al. (2021), who examine the labor market dynamics between 2012 and 2020, during the Abenomics period.

In comparison, younger women exhibit similar trend, with agriculture decreasing from 3.0% to 1.3%. The share of the services sector remained unchanged at 33.6%, while the share of the medical/healthcare sector grew from 15.6% to 22.8%. The overall trend among younger women mirrors that of older women but with more subtle changes across industries.

3 Model

In this section, we present our quantitative model. The unit of the model is an individual. We consider a life-cycle model of individuals who are heterogeneous in multiple dimensions. We focus on male individuals and abstract from heterogeneity in the family structure. Individuals differ in age, assets, health status, labor productivity in both fixed and stochastic components. We also assume preference heterogeneity in patience to account for a fraction of individuals who save none or very little over the life-cycle.

In each period, individuals choose the amount of assets to carry to the next period, and decide whether to participate in the labor market, and if so, how many hours to work. Individuals face uncertainty in three dimensions. First, the life-span is uncertain and they face survival probabilities in each period that vary by age. Second, health status is uncertain, which we assume affects utility cost of participating in the labor market. Third, labor productivity includes an idiosyncratic stochastic component. We assume that individuals take the macroeconomic conditions, including factor prices and fiscal policies and there is no aggregate shock.

3.1 Demographics

Individuals enter the economy at age $j = 1$ and live up to the maximum age of $j = J$. Lifespan is uncertain and the conditional probability of survival from age j to age $j + 1$ is denoted as s_{j+1}, with $s_{J+1} = 0$.

3.2 Endowments

In each period, individuals are endowed with a unit of disposable time to market work and leisure. Individuals productivity consists of three components. First, individuals are born with ability type α, which is fixed throughout their life-cycle. Second, there is an idiosyncratic stochastic component denoted by η, which we assume follows a finite-state Markov process. Third, there is age-specific productivity denoted as ν_j. Earnings of a working individual is given as $y = \alpha\eta\nu_j hw$, where h denotes work hours and w is the wage per efficiency unit. The interest rate r is exogenously given.

3.3 Health

Health status of an individual is denoted as m and $\pi_j^m(m, m')$ represents the probability that the health status is m' in the next period when the current health status is m, for an individual of age j.

3.4 Preferences

Individuals derive utility from consumption c and leisure l in each period according to the following utility function.

$$u(c, l) = \frac{[c^\gamma l^{1-\gamma}]^{1-\sigma}}{1-\sigma},$$

where

$$l = L - h - (\theta_j + \phi_m)I_h.$$

L denotes the total time endowment of an individual, and h is work hours chosen by an individual. θ_j denotes participation costs that may vary with age, and ϕ_m represents participation costs that vary by health status. I_h is an indicator that takes a value of 1 if an individual participate, i.e. $h > 0$.

The lifetime utility of an individual is defined as

$$E\left\{\sum_{j=1}^{J} \beta_i^{j-1} u(c_j, l_j)\right\}, \tag{2}$$

where the expectation is with respect to idiosyncratick producitivity shocks, mortality risks and health shocks. We assume preference heterogeneity in discount factors. β_i in

equation (2) denotes subjective discount factor of type-i individuals and we assume the value remains constant throughout the life-cycle.[7]

3.5 Government

The government collects taxes from households and provides social security benefits and means-tested transfers.

Once reaching the pension eligibility age, or normal retirement age j_R, each individual starts to receive pension benefits $ss(p)$. The benefits are given as an increasing function of past earnings. They depend on p, an index that summarizes an individual's past earnings up to the normal retirement age, which are subject to a proportional payroll tax denoted as τ^p.

The index p evolves as

$$p' = \frac{(j-1)p + y}{j}$$

for $j < j^R$ and $p' = p$ for $j \geq j^R$, where y is an individual's earnings at age j as discussed above. Note that the normal retirement age j_R is the age at which individuals start to receive public pensions benefits. The retirement age for the public pension does not have to be the same as the age at which individuals leave the labor force and retire from work.

Besides the payroll tax, the government imposes taxes on consumption at rate τ^c, labor income at τ^l, and capital income τ^a. The after-tax gross interest rate is denoted as $R = 1 + r(1 - \tau^a)$.

The government provides a transfer tr to an individual to guarantee a minimum consumption level of $\underline{c}$. The transfer amount is given as (3) and it is zero if disposable assets exceed $\underline{c}$ plus consumption taxes.

$$tr = \max\{0, (1+\tau^c)\underline{c} - \underline{a}\} \tag{3}$$

$\underline{a}$ denotes the disposable assets of an individual defined as

$$\underline{a} = Ra + (1 - \tau^l - \tau^p)y + ss$$

3.6 Individuals' Problem

Individuals are heterogeneous in four dimensions, and the state vector is given as $(j, a, i, \alpha, \eta, m, p)$, where j denotes age, a assets, i preference type, m health (medical) status and p an index for past earnings.

$$V(j, a, i, \alpha, \eta, m, p) = \max_{c,e,a'} \{u(c, l) + \beta_i s_{j+1} \mathbb{E}V(j+1, a', i, \alpha, \eta', m', p')\}$$

[7] See Krusell and Smith (1998) and Hendricks (2007), for example, for papers that incorporate heterogeneity in discount factors.

subject to

$$
\begin{aligned}
(1+\tau^c)c + a' &= Ra + (1-\tau^l-\tau^p)y + ss(p) + tr \\
y &= \alpha\eta\nu_j hw \\
a &\geq 0
\end{aligned}
$$

The expectation is with respect to the health status m' and productivity shock η' in the next period. Social security benefits, denoted as $ss(p)$, are paid to those at and above the normal retirement age j^R and it takes a value of zero otherwise. Welfare transfer tr is positive for eligible individuals and zero otherwise.

4 Calibration

In this section, we presents parametrization of the model. We use various database including the Labour Force Survey (LFS), the Basic Survey of Wage Structure (BSWS), the Japan Household Panel Survey (JHPS/KHPS), and estimates of the National Institute of Population and Social Security Research (IPSS), to obtain the data counter part of our model variables and to derive target moments of our calibration.

Our basic strategy is to calibrate parameters to match the life-cycle profiles of variables for the cohort born in 1936-1940, which we call as the 1940 cohort. This cohort is at age 80 by 2020 and we have labor market data for most of their working ages. We also use the demographic variables and government policies that vary over time, consistently with what this cohort experienced over the life-cycle. Table 3 summarizes the calibration results.

4.1 Demographics

Individuals enter the economy at age $j = 1$ (25 years old) and live up to the maximum age of $J = 76$ (100 years old). Survival probabilities s_j for the 1940 cohort is obtained from the Life Table reported by the Ministry of Health, Labour and Welfare (MHLW).

4.2 Endowments

An individual's earnings is given by $y = \alpha\eta\nu_j hw$ and the logarithm of the wage is given as

$$\log\alpha + \log\eta + \log\nu_j + \log w$$

We calibrate the distribution of the fixed ability type α and the stochastic process of the idiosycratic component η to match the life-cycle profile of wage dispersion estimated by Lise et al. (2014). They use the BSWS data to estimate various moments of male wages by age for different cohorts and we use their estimates for the 1940s cohort.

We assume that there are two values of α, a fixed ability type to match the variance of wages at the initial age of 25. Lise et al. (2014) demonstrate that the wage variance rises monotonically and linearly in age. To approximate this pattern, we assume that the idiosyncratic shock η are permanent and calibrate the variance of the error ε_η^2 to match the growth rate of the variance between age 25 and 59. The age-specific component ν_j is based on the average wage by age for individuals aged 25-59 from the BSWS.

The wage rate declines from age 60 onwards and we assume that the the age-specific component falls at rate δ from age 60 and set this value to match the ratio of average wage between 60 and 80 in the BSWS. Note that the depreciation rate δ is set in the model so that the growth rate of the average wage of participating workers matches the data. The interest rate is exogenous in the model and set to 2%.

4.3 Health

We assume that an individual's health status evolves stochastically over the life-cycle and affects participation cost that also varies by age. We use the JHPS/KHPS data to compute the health transition matrix $\pi_j^m(m, m')$. As we discussed in section 2, we use answers to the survey question about individuals' subjective health status and group them into three categories of good, fair and poor.

4.4 Preferences

In the utility function, we set γ, the weight parameter on consumption relative to leisure, so that the average work hours.[8] We set the risk aversion parameter σ to 2.

We allow for heterogeneity in the discount factor β_i, assume that there are two types to approximate the pattern of savings. There is a fraction of individuals who save very little over the life-cycle. In the JHPS/KHPS data, about 15% of samples report that they own zero asset around the retirement age. This is similar to what is reported in Kitao and Yamada (2019), who use the NSFIE data. We assume that 15% of individuals are hand-to-mouth (type 2), by setting β_2 to zero. For other individuals (type 1), we set the discount factor β_1 so that the average wealth at age 60 as in the data.

There are two types of participation cost, the component that depends on age θ_j and on health ϕ_m, as presented in section 3. For the age-dependent participation cost θ_j, we assume a functional form as follows and set the parameters Θ_1, Θ_2 and Θ_3 to match the

[8] Average weekly work hours for men aged 25-59 is 199 hours for the 1940 cohort, from the BSWS. We set the target $h = 199/480 = 0.42$.

participation profile of those aged 60 and above.[9]

$$\theta_j = \Theta_1 + \Theta_2 j^{\Theta_3}$$

Among the elderly aged 60-74, the average participation rate of those in fair and poor health is lower than that of those in good health by 10 and 37 percentage points, respectively. We set the participation costs for individuals in fair and poor health to match these differences in the model.

4.5 Government

We set the capital income tax rate τ^a and labor income tax rate τ^l to 0.35 and 0.18, respectively, based on the estimates of effective tax rates.

Consumption tax was first introduced in 1989 and set at 3% and raised gradually to reach 10% in 2019. We set the consumption tax rate τ^c accordingly.

Payroll tax captures the social insurance premium, including the payment for public pension, health insurance and long-term care insurance programs. The total tax rate paid by employed workers rose from 5.9% in 1965, when the 1940 cohort was 25 years old, to 15.0% in 2024. We let the payroll tax rate τ^p vary correspondingly.

The replacement rate κ is set so that the average pension benefits of the 1940 cohort matches with the data. We set the pension retirement age to 60 years old, $j_R = 41$. The retirement age was raised gradually since 2001, from 60 to 65, and we will simulate the model with an alternative retirement age and replacement rate in section 5.

[9] In the calibration algorithm, we use the average participation rate in the 60s and 70s, as well as the zero participation in the 90s as the target moments. The age in the functional form represents the number of years from age 59.

Table 3: Parameters of the Model

Parameter	Description	Value/Source
Demographics		
J	Maximum age	76 (age 100)
s_j	Survival probability	Life Table
Endowment		
α	Fixed ability	{0.77, 1.30}, Lise et al. (2014)
σ_η	Idio. productivity std. dev.	0.091, Lise et al. (2014)
ν_j	Age-specific productivity	See text, BSWS
δ	Skill depreciation rate	0.037, BSWS
Health		
$\pi(h, h')$	Health status transition	See text, JHPS/KHPS
Preference		
β_i	Subjective discount factor	{0, 0.998}
σ	Risk aversion parameter	2.0
γ	Preference weight	0.403
θ_j	Participation cost by age	See text, BSWS
ϕ_m	Participation cost by health	{0,0.015,0.064}, JHPS/KHPS
Government		
j_R	Pension retirement age	46 (age 65)
κ	Public pension replacement rate	0.417
τ_l	Labor income tax rate	0.18
τ_a	Capital income tax rate	0.35
τ_p	Payroll tax rate	Time-varying
τ_c	Consumption tax rate	Time-varying
Other Parameters		
r	Interest rate	2.0%

5 Numerical Results

5.1 Baseline Model

Figure 12 shows the employment rates of individuals in our baseline model, compared to the data for the cohort of men born in 1936-1940, constructed from the LFS data. Individuals in our model continue to work until age 60 and start to leave the labor force gradually, reaching nearly a full retirement by age 90, as in the data.

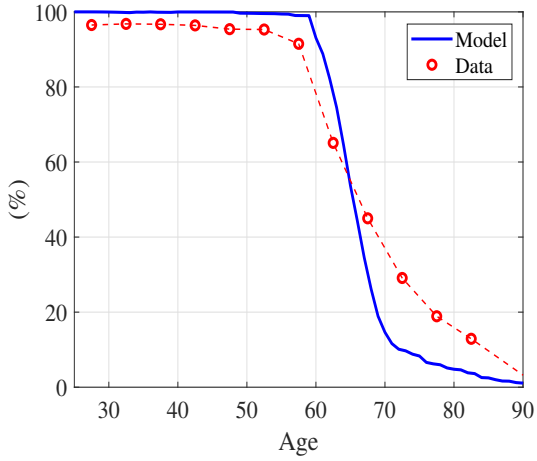


Figure 12: Employment Rates: Baseline Model

Note that we are not able to explain a few percentages of individuals aged between 25 and 60 who do not work in the labor market. It likely includes individuals who are not working for reasons such as unemployment, disability or hospitalization, etc, which are not accounted for by the elements of our model. The goal of our quantitative model is to approximate the pattern of gradual decline in employment for individuals above age 60 and our model, and we do so with a parsimonious set of parameters that define work and participation disutility and productivity.

Figure 13 shows the participation rates by health status for individuals aged above 60. Note that we match the difference in participation using the health-specific participation cost for individuals in fair and poor health status, respectively.

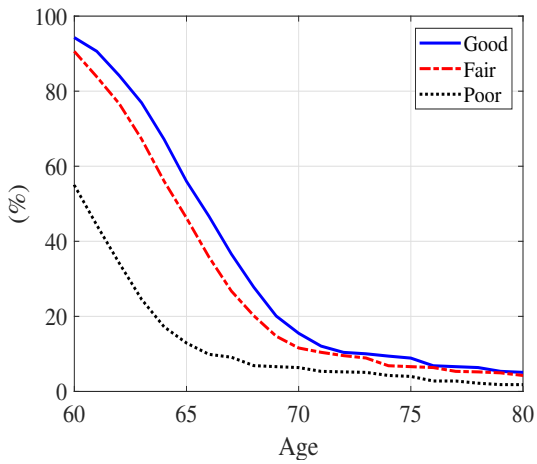
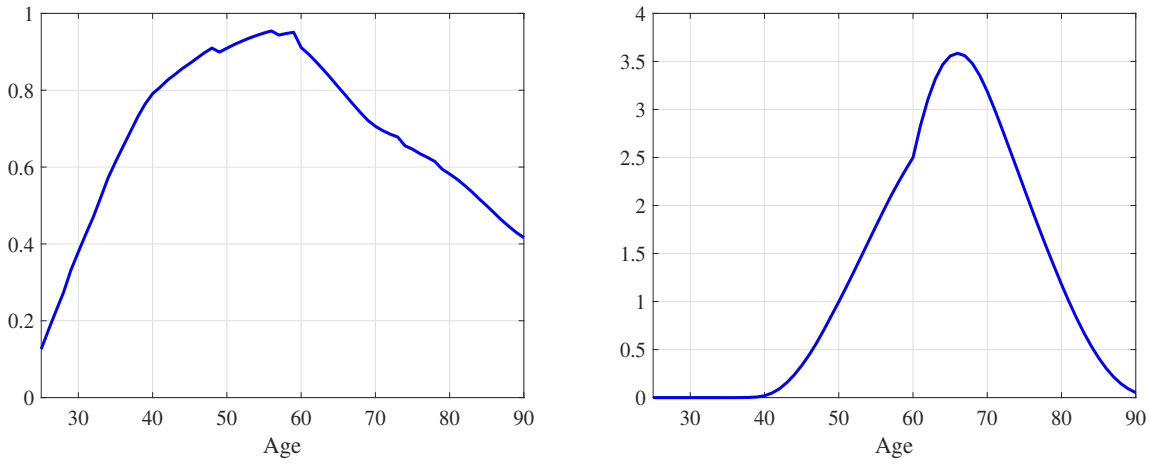


Figure 13: Employment Rates by Health (Age 60 and above): Baseline Model

Figure 14 shows the life-cycle profiles of consumption and assets. The consumption profile is hump-shaped, and in a similar shape as the estimates reported in the empirical literature, such as Kitao and Yamada (2024). The assets increase towards the retirement period and peaks in their mid-60s.

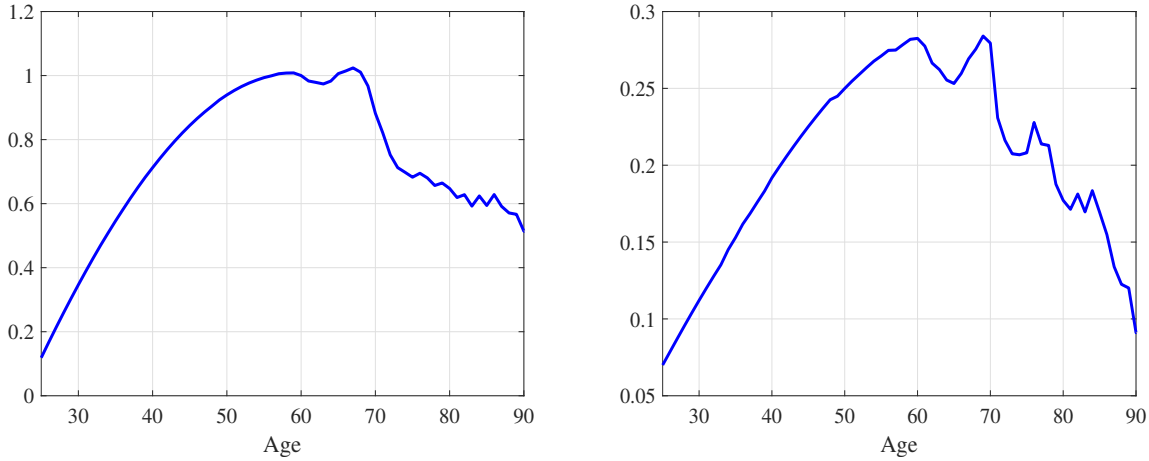


(a) Consumption

(b) Assets

Figure 14: Consumption and Asset over the Life-cycle: Baseline Model (Average Earnings=1)

Figure 15 shows the age profiles of average wage and wage variance. Both the level and variance rise over the life-cycle until around age 60 and decline thereafter.



(a) Average Wage (Age 60=1)

(b) Log-wage Variance

Figure 15: Average Wage Level and Variance over the Life-cycle: Baseline Model

Figure 16 shows average work hours of workers by age in the baseline model, expressed in the model unit and as a fraction of disposable time. They work approximately 40% of their disposable time until their late 50s and many of them start work fewer hours after 60, when they start to receive social security benefits. Workers productivity declines at old ages and deterioration of health also increase participation costs, which also contribute to a decline in work hours at old ages.

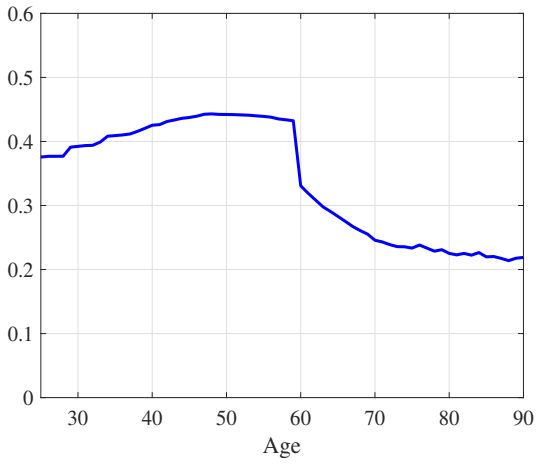


Figure 16: Average Work Hours of Workers: Baseline Model

Figure 17 shows the share of recipients of welfare transfers by age. According to the National Survey on Public Assistance Recipients of the Ministry of Health, Labour, and Welfare (MLHW), the share of welfare recipients rise in age and the profile is similar to that of our model.[10]

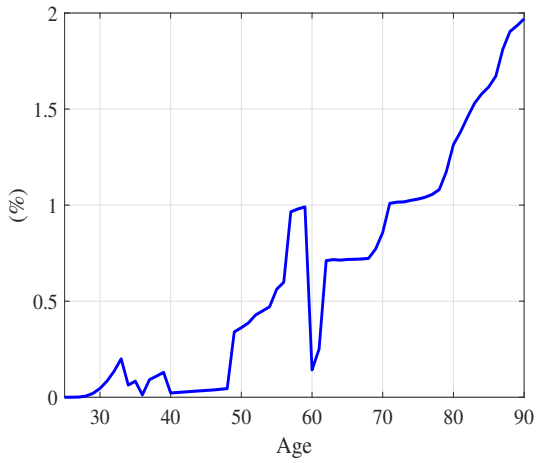


Figure 17: The Share of Welfare Recipients: Baseline Model

5.2 Factors Related to Elderly Participation

In section 2.4, we studied several factors that likely influence participation of the elderly. We now investigate how each of these and other factors affect life-cycle behavior of individuals in our model. In the first set of simulations, we assume alternative scenarios about the public pension scheme. In the second set of simulations, we consider alternative scenarios about the labor productivity process.

[10]Following the coverage by age for the 1940 cohort, the share increases from 0.6% in their 20s (in 1960), to 1.6% in their 60s (2000). Survey data is available at https://www.mhlw.go.jp/toukei/list/74-16.html (in Japanease).

5.2.1 Social Security Reform

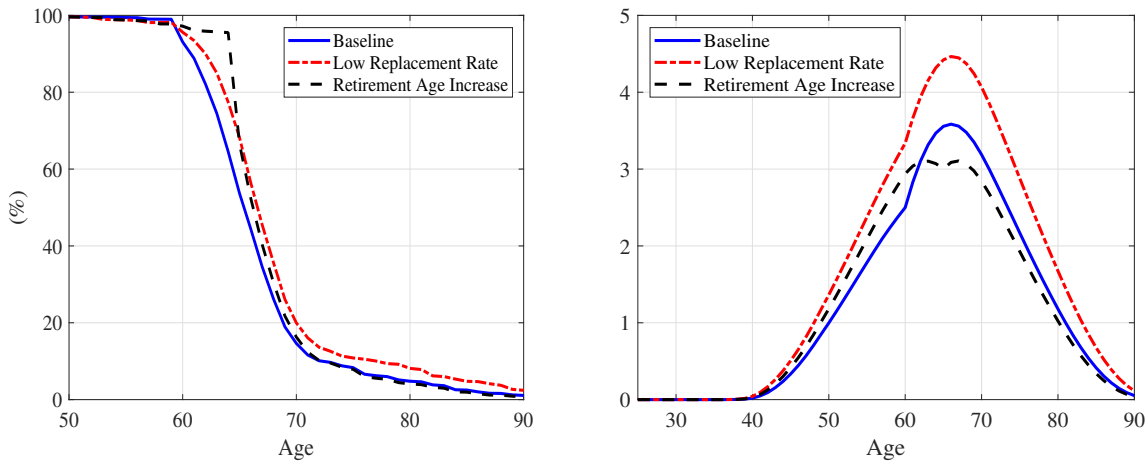
In the baseline model calibrated to the 1940 cohort, we assumed that individuals start receiving the public pension at age 60 and we had set the replacement rate to match the average pension benefits of the cohort. In this section, We simulate the economy under alternative assumptions about the normal retirement age and the replacement rate.

Since 2001, the retirement age has been raised gradually from 60, and it will be 65 by 2025 for men and by 2030 for women. The government also introduced the "macroeconomic slide" scheme in the pension reform in 2004. The scheme is supposed to adjust the amount of public pension automatically to the demographic changes; an extension of average life expectancy, and the decline in the number of the insured working-age population. The scheme sets an exemption from the adjustment under a deflationary economy to avoid a decline in nominal benefits. There were no adjustments due to this exemption and other reasons in many years and the adjustment was implemented only several times. According to the government's Financial Projection of Pension (*zaisei kensho*) of 2024, published by the Ministry of Health, Labour and Welfare (MHLW), the pension replacement rate will decline by approximately 20% under the baseline growth scenario.

We simulate a reform of raising the retirement age from age 60 to 65, and a reform to reduce the replacement rate by 20%. Figure 18 shows the paths of employment rates and assets under these two scenarios. With a low replacement rate, individuals have stronger incentives to save for retirement and the asset level is higher throughout the life-cycle. At age 65, the average asset will be 24% above that in the baseline model. Individuals also stay in the labor force longer and the average employment rate in their 60s is 67.3%, while it is 58.1% in the baseline model.

When the retirement age to start receiving pension benefits is raised from 60 to 65 years old, individuals save more than in the baseline model until the early 60s. Thereafter, they decumulate wealth faster as they need to wait for five more years until pension benefits are paid. Many individuals choose to stay in the labor force longer and the employment rate remains especially high until the new retirement age. The average employment rate in their 60s is 69.2%, about 11 percentage points above the baseline level.

The two experiments show that individuals prepare for the lower pension benefits by the combination of more savings for retirement and a longer working period. Combining the two scenarios, raising the retirement age to 65 and reducing the replacement rate by 20%, the average employment in their 60s will be 74.9%, 16.8 percentage points above the employment rate in the baseline model, indicating that part of the rise in the participation rate of the elderly in recent years is likely due to the changes in the social security system.



(a) Employment Rates

(b) Assets

Figure 18: Social Security Reform

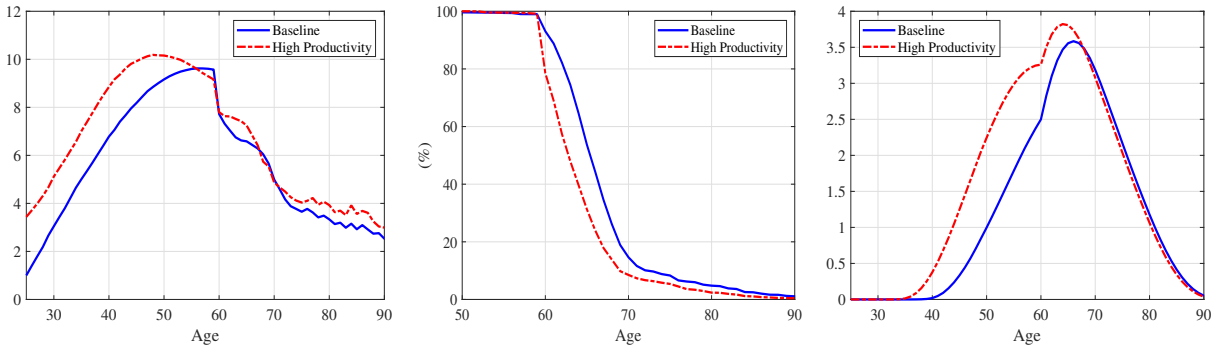
Note: Assets are expressed in units of average earnings in the baseline model.

5.2.2 Productivity Growth and Depreciation

The labor productivity of workers depend on age-dependent deterministic component ν_j, as well as other idiosyncratic components. We calibrated the deterministic component for individuals aged 25 to 59 so the wage profile by age would match the profile of the 1940 cohort. After age 60, we assume that the component will decline at rate δ, which we calibrated in the model to match the wage decline above age 60.

In this section, we simulate the model with two alternative scenarios about individuals' labor productivity. In the first, we assume higher average productivity of prime-age individuals aged between 25 and 59, using the wage profile of the 1960 cohort based on the BSWS data. In the second, we consider a scenario with a lower depreciation rate of old individuals' productivity, by setting δ at one percentage point below the baseline level.

Figure 19 shows the results for the scenario of high productivity. As shown in Figure 19a, labor income is much higher among the young and middle-aged individuals and they accumulate more wealth. The asset level at the retirement age of 60 is 30.3% higher than in the baseline model. With the strong income effect, employment rate is lower after the retirement age, as shown in Figure 19b.

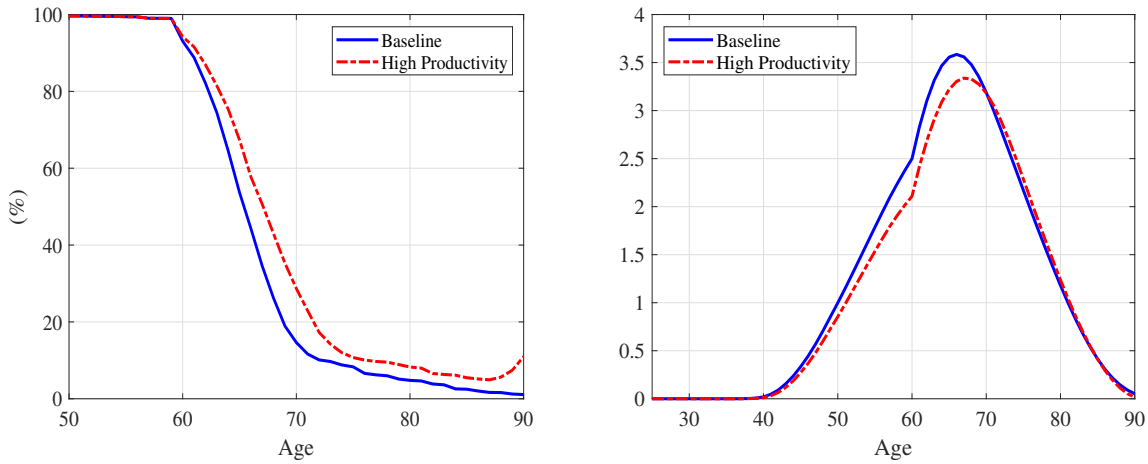


(a) Earnings (b) Employment Rates (c) Assets

Figure 19: Productivity Growth

Note: Average earnings are for workers only. Earnings and assets are expressed in units of average earnings in the baseline model.

Figure 20 shows the profiles of employment rates and asset profile, when we assume a lower depreciation rate of skills after age 60. Employment rates will decline more mildly after the retirement age and the average employment rate in their 60s is 68.4%, approximately 10 percentage points higher than in the baseline model. Anticipating higher wages and longer working years, individuals save less than in the baseline model until around age 70, as shown in Figure 20b, although the asset level is similar thereafter as their earnings are higher on average.



(a) Employment Rates (b) Assets

Figure 20: Higher Productivity of Old Workers

Note: Assets are expressed in units of average earnings in the baseline model.

The two experiments demonstrate that higher productivity and wages may or may not lead to more employment of the elderly. Economic growth and higher income in general imply strong income effects and reduce participation, but a rise in productivity of the elderly will induce them to work longer. Changes in the working capacity of

the elderly or changes in the labor market and working conditions of the elderly could be achieved in different ways, for example, by improvement of health conditions and working environment, better maintenance of human capital through recurrent and life-long education, and a shift towards jobs that require less physical strength.

6 Conclusion

This paper consists of two parts and in the first part, we provide a descriptive empirical analysis of labor market trends among the elderly in Japan. We then construct a quantitative structural model calibrated to the life-cycle behavior of the cohort of individuals born between 1936 and 1940.

Using data from the Labour Force Survey (LFS) and the Basic Survey of Wage Structure (BSWS), we analyze trends in employment, earnings and work hours for men and women over the past several decades. Participation rates for those aged 60 and above are significantly higher in Japan than in many other OECD countries, and these rates have continued to rise since 2000. However, average earnings, however, of young and middle-aged male workers have been declining since the 1990s. Additionally, we use data from the Japan Household Panel Survey (JHPS/KHPS) to study factors related to the elderly's employment.

We construct a model of heterogeneous individuals to account for their life-cycle behavior, allowing for heterogeneity in age, labor productivity, assets, health status, preference in discount factor, and past earnings. The calibrated model is used to quantify the effects of factors related to elderly participation decisions. We find that pension reforms to lower the replacement rate or raise the retirement age lead to higher participation rates among the elderly. Additionally, we show that an overall productivity increase and higher wages of young and middle-aged workers will lower participation of elderly due to income effects, but higher productivity of the elderly due to lower skill depreciation provides more incentives to stay in the labor force.

Finally, we discuss several limitations of our quantitative analysis and suggest areas for future research. First, as a partial equilibrium analysis, our model does not consider how changes in individual behavior might affect macroeconomic variables, such as factor prices, tax revenues and government expenditures. Exploring these general equilibrium effects under alternative policy scenarios owuld be a valuable extension. Second, our structural model focuses on male behavior, while the empirical section highlights that employment and earnings patterns differ between men and women, with women experiencing different trajectories over the past decades. Developing a model to account for both men and women's behavior, possibly considering heterogeneity in family structures, would be a challenging but important direction for future research.

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